DOD-C-64018(AR)
03 OCTOBER 1983

MILITARY SPECIFICATION

CARTRIDGE, 30MM, HIGH PRESSURE TEST, HP-T, M883

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electrically primed cartridge for use in the proof testing of 30MM guns and barrels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Lot Numbering of Ammunition

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. Army Armament Research & Development Center (ARDC)

Product and Packaging Drawings

11825941	-	Cartridge, 30MM HP-T, M883
11825892	-	Linked Packing - Left Hand
11825891	-	Linked Packing - Right Hand
11825890	-	Bulk Packing
10542565	-	Shipping and Storage Container M592
11825879	-	Link
11825901	-	Case Cartridge Assembly

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Code of Federal Regulations

Title 49	-	Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 General.

3.1.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg. 11825941), all associated drawings, and with all requirements specified in applicable specifications.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the Contracting Officer, and no deviation from that process shall be made without prior approval (ref 6.3).

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.4 Projectile extraction. The projectile, when subjected to a bullet pull using a chamber simulating ring at a rate of 25MM per minute + 10 percent, shall have a retention force such that the mean value less three standard deviations shall be greater than 8 kilo Newtons.

3.5 Electrical resistance, primed case. The electrical resistance of the primed case shall be as specified on Dwg. 11825901.

3.6 Propellant charge. The propellant charge used in each cartridge shall be maintained to meet the ballistic requirements.

3.7 Chamber pressure. When fired at $21^{o} \pm 5^{o}$ C the peak chamber pressure shall be 355 MPa minimum.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Process defects. The cartridges shall be free of defects, such as, folds, wrinkles, buckles, splits, severe scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.8.2 Cleanliness. The cartridge and its components shall be free of all foreign matter (see 6.9).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-64018(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
	Cartridge, 30MM, HPT, M883				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case Assembly - Dwg. 11825901 Examination for defects	32		3.1.2	4.4.2.1
	Cartridge, 30MM, HPT, M883 Projectile extraction	32	ACC-1 REJ-2	3.4	4.5.5
	Propellant charge	32	ACC-0 REJ-1	3.6	4.5.3
	Pressure	32	ACC-0 REJ-1	3.7	4.5.2

NOTES:

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Cartridge cases from one interfix number from one manufacturer.

b. Projectile parts from one interfix lot number from one manufacturer.

c. Primers from one interfix lot number from one manufacturer.

d. Propellant from one lot number from one manufacturer.

e. Loaded cartridge case assemblies from one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-64018(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Case Cartridge Assembly 1/				11825901
					NEXT HIGHER ASSEMBLY 11825941
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Depth of primer seating	4.4.2a	.25%	3.1.2	Gage
102.	Primer crimp missing		.25%	3.1.2	Visual
103.	Primer/case resistance		.25%	3.1.2	Gage
104.	Presence of igniter tape seal		.25%	3.1.2	Gage
105.	Dimpling of cartridge case		.25%	3.1.2	Visual
MINOR					
201.	Waterproof missing around primer	4.4.2a	.65%	3.1.2	Visual
202.	Incomplete or missing primer crimp		.65%	3.1.2	Visual
203.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:
See notes following 4.4.2.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-64018(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Cartridge, 30MM, HPT, M883 1/				11825941
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Split or perforated case	4.4.2a	100% 2/	3.1.2	Visual
2.	Crack or split in projectile		100% 2/	3.1.2	Visual
MAJOR					
101.	Cocked, loose, missing or inverted primer	4.4.2a	.25%	3.1.2	Visual
102.	Overall length, max.		.25%	3.1.2	Gage
103.	Improper or incomplete crimp		.25%	3.1.2	Visual
104.	Case crimp diameter (max.) 3/		.25%	3.1.2	Gage
105.	Propellant charge (min.) 4/		.25%	3.1.2	Weight
106.	Max. gap case projectile interface		.25%	3.1.2	Gage
107.	Profile and alignment		.25%	3.1.2	Gage
MINOR					
201.	Rotating band damaged	4.4.2a	.65%	3.1.2	Visual
202.	Deformed nose		.65%	3.1.2	Visual
203.	Protective coating damaged or incomplete 5/		.65%	3.1.2	Visual
204.	Incorrect type cartridge or components		.65%	3.1.2	Visual

NOTES:
See notes following 4.4.2.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-64018 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge, 30MM, HPT, PXR6317 1/	2 OF 2	11825941
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205.	Marking incorrect, incomplete, illegible or missing		.65%	3.1.2	Visual
206.	Scratch, dent, buckle, bulge, wrinkle or fold in case		.65%	3.1.2	Visual
207.	Evidence of poor workmanship		.65%	3.1.2	Visual
	TEST				
	Projectile extraction	20	ACC-0 REJ-1	3.4	4.5.5
	Propellant charge	20	ACC-0 REJ-1	3.6	4.5.3
	Pressure, min	20		3.7	4.5.2

NOTES:

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NOTES FOR PARAGRAPH 4.4.2.1 & 4.4.2.2

1/ Visual standards/machine masters established and approved by the Government as defined in 6.4, shall be used in evaluating cartridge case and cartridge visual defects. Components shall have previously been accepted in accordance with the acceptance criteria defined in the specification listed in the applicable drawing.

2/ Inspection verification of prior 100% inspections shall be performed per (a) by utilization of Government approved machine masters or (b) MIL-STD-105, AQL = 0.25, to determine sample size, minimum sample size = 315, zero defects allowed.

3/ The lot shall not be penalized during inspection verification for oversize conditions provided that:

a. The crimp is visually complete for 360° and,

b. the projectile extraction requirement of 3.4 are met.

4/ A cartridge weighing less than the value determined by the formula specificed in 6.8 shall be rejected. The 100% inspection method may be by verification of weight of each cartridge or by probe for a fixed minimum propellant height in a charged cartridge case. If the probe method is used, the following method shall be used to establish the minimum height for each cartridge lot.

a. Establish the mean charge weight in accordance with 4.5.3.

b. Charge a minimum of 25 cases to the specified weight, measure each case and calculate the mean propellant height.

c. Repeat step b with a propellant charge weight 1 gram less than the established mean charge weight.

d. Adjust probe height such that any cartridge in which the propellant charge is 1 gram below the mean weight will be rejected.

e. Repeat steps a through d for each propellent batch and cartridge case lot serial number.

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5/ The lot shall not be penalized for anodize crazing in the cartridge case crimp grooves (see 6.5).

4.5 Test methods and procedures.

4.5.1 Unlisted firing defects. The lot should be suspended and referred to the contracting officer for disposition if any malfunction or casualty not covered by this specification occurs in any firing test.

4.5.2. Chamber pressure. The test rounds shall be conditioned at 21°C and fired from a test fixture barrel. A permanent record shall be made of the resulting pressure. The record may be a pressure trace, paper print out of the data or other similar instrument reading. The pressure shall conform to the requirements of 3.7.

4.5.3 Propellant charge. The charge of propellant from a given lot is required to meet the pressure requirement of 3.6 The minimum charge weight assigned shall be used in determining the minimum cartridge weight in accordance with 6.8.

4.5.4 Primed cartridge cases. The primed cartridge case resistance shall meet the requirements of drawing 11825901.

4.5.5 Projectile extraction. The base of the cartridge case shall be secured in a holding fixture and a ring simulating the gun chamber shall be installed in the case neck area. A clamp shall be applied along the longitudinal axis of the round. An axial load shall be applied at a constant strain rate of 25MM per minute ± 10% until the projectile is withdrawn from the cartridge case. The peak axial load shall be recorded and compared to the requirements of 3.4. A case neck separation during this test shall be cause for rejection. A split of greater than 5MM observed during projectile extraction shall be cause for rejection.

4.5.6 Inspection equipment. The inspection equipment required to performed the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.9 herein.

5. PACKAGING

5.1 Packaging and packing. The cartridge shall be preserved, packaged and packed in accordance with the applicable requirements for the shipment of hazardous items as defined in Title 49, Code of Federal Regulations, Parts 100 through 199.

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5.1.1 Packing.

5.1.1.1 Linked pack. The linked pack shall consist of linked cartridges packed in the M592 shipping and storage container, 10542565 and may include provisions for interconnecting ammunition belts. The links shall conform to Drawing 11825879. The pack shall also include provisions for a right and left hand pack. The packing ratio of left to right belts shall be specified in the contract or purchase order.

5.1.1.2 Bulk packing. The bulk pack shall consist of unlinked cartridges packed in the M592 shipping and storage container.

5.1.2 Ammunition container interior packing.

5.1.2.1 Linked packing. The interior packing shall be in accordance with Drawing 11825891 or 11825892 as applicable.

5.1.2.2 Bulk packing. The interior packing shall be in accordance with Drawing 11825890.

5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with Dwg. No. 11825893. In addition, exterior containers shall show the caution makings required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition.

5.3 Documentation with shipment. Each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with MIL-STD-1168, date of manufacture and complete definition of the ammunition including traceability of such items as case, primer, flashtube assembly, propellant, projectile and the charge weight.

5.4 Shipping. When shipments of more than one lot are made at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This specification covers assembling and packing for Cartridge, 30MM, High Pressure Test (HPT) M883 for use in the proof testing of 30MM guns and barrels.

6.2 Ordering data. See MIL-A-48078.

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6.3 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.4 Visual defects standards. Components and assemblies which have been rejected during initial production shall be segregated and visual standards established jointly by representatives of the Government and the contractor. Subsequent application of these standards for acceptance inspection shall be specified in the contract or purchase order.

6.5 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pre-treatment shall be considered unsatisfactory if any of the following conditions exist.

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters.

c. More than five areas of 15 square millimeters or less.

6.6 Rotating band failure criteria. Rotating band failure shall be defined as loss of any portion of the band which exceeds a 13MM length and has evidence of band seat serration imprint, or the wiping down to the bourellet diameter (no engraving in evidence) over the total length of the rotating band. For classification as a defect, there must be evidence, either by microflash photographs, recovery of the band or portions thereof, or by hole (s) in the witness board/screen. Normal band fringing is acceptable.

6.7 Data reporting. Final data calculation for pressure shall be rounded off to the nearest 0.1 MPa.

6.8 Minimum cartridge weight. The minimum cartridge weight shall be determined for each lot based on the minimum propellant charge weight selected per 4.5.3. The formula for minimum cartridge weight, when assembled in accordance with Drawing 11825947 is defined as follows:

Minimum weight (grams) = 292 grams + charge weight (grams)

6.9 Contamination of explosive components. Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s).

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6.10 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAF-I(D), Dover, NJ, 07801. The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

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Preparing Activity:
Army-AR

Project No. 1305-A912

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-64018(AR)	Cartridge, 30MM , High Pressure Test, HP-T, M883

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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